

# Holden Swindell

|             |                                                                                                       |
|-------------|-------------------------------------------------------------------------------------------------------|
| Email       | swind027@umn.edu                                                                                      |
| Website     | <a href="https://hswindell.github.io">https://hswindell.github.io</a>                                 |
| ORCID       | 0009-0002-9619-9386  |
| Institution | University of Minnesota Twin Cities                                                                   |
| Office      | Vincent Hall 422                                                                                      |

## Education

---

|                                                                       |                       |
|-----------------------------------------------------------------------|-----------------------|
| <b>University of Minnesota Twin Cities</b><br><i>Mathematics, PhD</i> | August 2025 – Present |
|-----------------------------------------------------------------------|-----------------------|

|                                                                                                      |                           |
|------------------------------------------------------------------------------------------------------|---------------------------|
| <b>University of Wisconsin-Madison</b><br><i>Mathematics, BSc and Computer Sciences, Certificate</i> | September 2021 – May 2025 |
|------------------------------------------------------------------------------------------------------|---------------------------|

## Research Experience

---

|                                                                                                                                                                                                                                                  |             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>UW-Madison REU in Analysis</b><br><i>Analysis and Partial Differential Equations RTG, University of Wisconsin-Madison</i><br>Website: <a href="https://web.math.wisc.edu/analysisrtg/node/7">https://web.math.wisc.edu/analysisrtg/node/7</a> | Summer 2023 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|

|                                                                                                                                                                                                                                                                             |           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>MXM Project “Random Symmetries of Hyperbolic Spaces”</b><br><i>Madison Experimental Mathematics Lab, University of Wisconsin-Madison</i><br>Website: <a href="https://mxm.math.wisc.edu/past-semesters/fall-2022">https://mxm.math.wisc.edu/past-semesters/fall-2022</a> | Fall 2022 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|

|                                                                                                                                                                                                                                                                |             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>MXM Project “Geometry of Random Integer Matrices”</b><br><i>Madison Experimental Mathematics Lab, University of Wisconsin-Madison</i><br>Website: <a href="https://mxm.math.wisc.edu/past-semesters/sp22">https://mxm.math.wisc.edu/past-semesters/sp22</a> | Spring 2022 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|

## Teaching Experience

---

|                                                                                    |           |
|------------------------------------------------------------------------------------|-----------|
| <b>Teaching Assistant</b><br><i>Math 1271, University of Minnesota Twin Cities</i> | Fall 2025 |
|------------------------------------------------------------------------------------|-----------|

|                   |                           |
|-------------------|---------------------------|
| <b>Lead Tutor</b> | Spring 2024 – Spring 2025 |
|-------------------|---------------------------|

|                                                                                                                                                                                        |                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>Tutor</b><br><i>Math Learning Center, University of Wisconsin-Madison</i><br>Website: <a href="https://math.wisc.edu/undergraduate/mlc">https://math.wisc.edu/undergraduate/mlc</a> | Spring 2022 – Fall 2023 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|

## Professional Memberships

---

|                                                                                                                                                              |                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>American Mathematical Society</b><br><i>Association for Professional Mathematicians</i><br>Website: <a href="https://www.ams.org">https://www.ams.org</a> | Fall 2025 – Present |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|

|                                                                                                                                            |                     |
|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Spectra</b><br><i>Association for LGBTQ+ Mathematicians</i><br>Website: <a href="https://www.lgbtmath.org">https://www.lgbtmath.org</a> | Fall 2024 – Present |
|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------|

## Service & Community Outreach

---

### Student Mentor

Fall 2025

*Saturday Morning Math, University of Minnesota Twin Cities*

Website: <https://cse.umn.edu/mathcep/saturday-morning-math>

### President

Fall 2024 – Spring 2025

### Vice President

Fall 2023 – Spring 2024

### Board Advisor

Fall 2022 – Spring 2023

*Math Club, University of Wisconsin-Madison*

Website: <https://win.wisc.edu/organization/mathclub>

### Directed Reading Mentor

Spring 2024

*Category Theory Reading Group, University of Wisconsin-Madison*

Website: <https://hswindell.github.io/category-theory.html>

## Honors & Awards

---

UW-MADISON

Honors in the Liberal Arts

Honors in the Major (Mathematics)

Phi Beta Kappa Honor Society Member

Dean's List (all Semesters)

## Talks

---

### “Equivalence via Surjections”

October 2025

*arXiv Seminar, University of Minnesota Twin Cities*

### “Point-Free Topology”

April 2024

*Category Theory Reading Group, University of Wisconsin-Madison*

### “Smooth Infinitesimal Analysis”

February 2023

*Math Club, University of Wisconsin-Madison*

## Conference Attendance

---

### Equivariant, Motivic, and Physical Topology in the Midwest

October 2025

*Conference in Honor of Igor Kriz*

Website: <https://z.umn.edu/igorfest>